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Prolonged cognitive effort impairs inhibitory control and causes significant mental fatigue after an endurance session with an auditive distractor in professional soccer players

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ABSTRACT

Background: Throughout official soccer matches, the presence of cheer by the crowd could be considered a critical auditive distraction that could further impair the cognitive interference control system, multiple object tracking (MOT) skill, heart rate variability (HRV), and increase mental fatigue. As the resource is not immediately replenished, the impairment of the cognitive interference control system may be delayed following a soccer game. Then, evaluating the recovery time course of the cognitive interference control system, MOT skill, HRV, and mental fatigue after prolonged tasks combining physical, endurance, and cognitive effort are essential.

Purpose: We aimed to analyze the acute effect of cognitive effort and auditive distractor with 24-h follow-up throughout a prolonged endurance session on inhibitory control, subjective mental fatigue, MOT skill, and HRV in professional soccer players.

Methods: Twenty professional male soccer players were recruited (23.56 ± 3.8 years, 78.1 ± 6.9 kg, 1.77 ± 0.06 m, and 12.5 ± 5.3% body fat). The sessions were performed in a randomized and counterbalanced crossover design, divided into four experimental conditions: endurance, endurance + MOT, endurance + MOT + AD, and endurance + AD. The soccer players completed the incongruent Stroop task utilizing an eye-tracker to assess cognitive effort. MOT task, subjective mental fatigue, and HRV were evaluated before the endurance training (60%Δ of maximal aerobic velocity during 40-min) and after 30-min and 24-h of recovery. These sessions were designed to investigate the acute effect of prolonged cognitive effort (repeated MOT throughout the endurance task) and AD (constant crowd noise and coach's voice each 15–40 s, totalizing = 80 voices) on inhibitory control, MOT skills, HRV, and subjective mental fatigue after a fixed endurance training session.

Results: There was no condition × time interaction for accuracy of inhibitory control ($p > 0.05$, $\eta p^2 = 0.001$). There was a significant condition × time interaction for inhibitory control response time ($p < 0.05$, $\eta p^2 = 0.16$). A higher response time of inhibitory control was found for the endurance + MOT + AD and endurance + MOT experimental sessions ($p < 0.05$). There was a significant condition × time interaction for subjective mental fatigue ($p < 0.05$, $\eta p^2 = 0.46$). A higher subjective mental fatigue was found for the endurance + MOT + AD and endurance + MOT experimental sessions ($p < 0.05$). There was no condition × time interaction for HRV ($p > 0.05$, $\eta p^2 = 0.02$).

Conclusion: We concluded that cognitive effort throughout a prolonged endurance session impaired inhibitory control and increased mental fatigue without promoting greater MOT skill and HRV changes in professional soccer players.

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1. Introduction

Prolonged cognitive activity of low or high cognitive demand might cause a psychobiological state called mental fatigue that presents symptoms such as tiredness, lack of energy, higher-than-normal perception of effort, and reduced cognitive ability, resulting in a performance reduction in a subsequent task (Brown et al., 2020; Jacquet et al., 2021; Pageaux et al., 2013; Smith et al., 2018). The negative effect of mental fatigue on physical performance is well-documented (Giboin & Wolff, 2019), affecting crucial aspects of sports performance such as endurance, tactical and technical, and executive function (Smith et al., 2018; Van-Cutsem et al., 2017). Then, cognitive effort control is crucial for success in high-level team sports once players must maintain endurance and perceptual-cognitive skill performance for prolonged periods during training (e.g., a 40-min endurance training with high visual attention) (Bigliassi, 2021; Faubert, 2013) and matches (e.g., a 90-min soccer match) (Fortes et al., 2021; Gantois et al., 2020). However, cognitive effort seems to impair the cognitive interference control system (André et al., 2019; Angius et al., 2022), reducing athletes' physical and cognitive performance.

The cognitive interference control system is critical for sports performance. In team sports athletes (i.e., soccer, basketball, and volleyball), the necessity to inhibit unwanted behavior or perceptual-cognitive information (e.g., anticipation, decision-making, and multiple object tracking) such as visual attention deviation (e.g., saccadic eye movements) are made by this interference control system (Romeas et al., 2016). Interestingly, frequent visual attention (e.g., multiple object tracking skill) to relevant stimuli throughout the training sessions (e.g., teammates and opponents) requires high cognitive activity and further higher cognitive effort (Fortes et al., 2021; Smith et al., 2016). In addition, the cognitive inhibition from the noises made by the crowd, cheerleaders, and coaches throughout the matches and training sessions also requires high cognitive effort (Englert et al., 2015; Furley et al., 2013; Gantois et al., 2020). During soccer training sessions, one of the most formidable brain tasks of a soccer player is to perceive and integrate complex moving patterns while allocating attentional resources to different key areas of dynamic scenes (Faubert & Sidebottom, 2012), inhibiting irrelevant auditive stimuli for the task. Furthermore, players and object movements (e.g., ball or puck) might be extremely fast and diverse, abruptly changing speed. In this sense, the multiple object tracking (MOT) skill is frequently used in soccer matches and training sessions due to the rapid change of direction of teammates, opponents, and ball (Romeas et al., 2016) while irrelevant auditive stimuli should be inhibited, increasing cognitive demands in soccer players (Englert et al., 2015; Furley et al., 2013; Smith et al., 2018).

A recent review of the neural basis of attentional focus throughout physical endurance tasks revealed an impaired cognitive interference control system throughout endurance tasks with the insertion of distractive stimuli (Bigliassi, 2021). For example, during soccer training sessions, journalists and fans could be considered a critical auditive distraction that could further impair the cognitive interference control system and increase mental fatigue. In addition, previous studies revealed an impaired cognitive interference control system with a concomitant reduction in heart rate variability (HRV) in mentally fatigued subjects (Qin et al., 2021; Tanaka et al., 2009). The HRV seems partially regulated by brain areas associated with the cognitive interference control system (Tanaka et al., 2012). Previous findings showed impaired physical endurance performance when the HRV was attenuated (Boullosa et al., 2021; Pereira et al., 2019). It is important to highlight that HRV is reduced during or after a high training load (Boullosa et al., 2021), such as prolonged cognitive effort plus physical endurance tasks (e.g., 40-min for physical endurance task with moderate intensity) with an auditive distractor. So, it seems crucial to measure internal training load after prolonged cognitive effort plus physical endurance task with auditive distractor to confirm this hypothesis. Once true, these findings can contribute to soccer coaches starting to control cognitive and physical load during training sessions.

The soccer training session is characterized by a combination of prolonged physical effort (Fortes et al., 2021) and increased cognitive load (Faubert & Sidebottom, 2012). The high cognitive load throughout a soccer training session could be explained by thousands of decision-making (i.e., motor and cognitive) (Gantois et al., 2020), prolonged visual attention for multiple relevant stimuli (i.e., MOT skill) (Romeas et al., 2016), and inhibition of irrelevant auditive stimuli for the task (e.g., noise from crowd and cheerleaders). Therefore, it is reasonable to claim that this overload could hamper the cognitive interference control system and cause mental fatigue, depleting cognitive resources (Baumeister et al., 2007). Thus, evaluating the time course of recovery of the cognitive interference control system and mental fatigue after prolonged tasks combining physical endurance and cognitive effort is essential for providing information about how long it takes for players to totally recovery from the last match or training session.

In the present study, an experimental, controlled, randomized, and mechanistic investigation was performed to understand the impact of a controlled soccer training session (e.g., endurance with moderate intensity and cognitive effort together per 40-min) with auditive distractor stimuli on important measures frequently used by soccer coaches to analyze the state of physiological (e.g., HRV) and cognitive (e.g., inhibitory control, mental fatigue level, and MOT skill) readiness of players. The endurance intensity was 60%Δ of maximal aerobic velocity ($\dot{V}$ max) performed on a motor-driven treadmill. The chosen exercise was intended to avoid early exhaustion throughout the 40-min whole-body endurance task. Previous studies revealed that with intensity at 70%Δ Vmax, the participants remained less than 30-min in the task (Midgley et al., 2007; Walsh et al., 2010). Also, the moderate intensity (i.e., 60%Δ of Vmax) was chosen to avoid greater cumulated neuromuscular fatigue throughout the experimental session so that it would not interfere with the experiment results. Although the endurance task does not precisely represent in-field physical demands, the whole-body endurance task was continuous (i.e., 40-min) to allow the MOT task to simulate the repeated visual attention and cognitive effort throughout a soccer training session.

The findings from the present scientific investigation will provide information on whether the prolonged cognitive effort (e.g., MOT) and endurance task plus auditive distractor together represent a stimulus that might impair inhibitory control, MOT skill, HRV, and increase subjective mental fatigue until 24-h after an endurance training session in professional soccer players. The follow-up measures (i.e., until 24-h) provide information about the recovery time of physiological and cognitive readiness measures after a simple soccer training session. This information could be important for technical staff and soccer coaches, suggesting whether a new training session may be inserted shortly after an endurance task with auditory distractor or whether the inhibitory control, considered important to answer interviews after a soccer training session, may be impaired after a soccer training session with auditive distractor.

Thus, we aimed to analyze the acute effect of cognitive effort and auditive distractor with 24-h follow-up throughout a prolonged endurance session on inhibitory control, subjective mental fatigue, MOT skill, and HRV in professional soccer players. For this proposal, Stroop task (i.e., inhibitory control), subjective mental fatigue, MOT skill, and HRV were measured immediately prior, 30-min, and 24-h after the experimental sessions. This investigation was designed with four experimental conditions that included endurance (i.e., 60%Δ of maximal aerobic velocity during 40-min), endurance and cognitive effort (i.e., repeated MOT skills during 40-min), endurance, cognitive effort, and auditive distractor (i.e., constant crowd noise and coach's voice), and endurance and auditive distractor. We hypothesized that cognitive effort with auditive distractor throughout prolonged endurance sessions impairs inhibitory control, MOT skill, HRV, and increases subjective mental fatigue until 24-h after an endurance training session in professional soccer players.

2. Materials and methods

2.1. Participants

We calculated the implied sample size power using an analysis of variance (ANOVA) with repeated measures within factors. The sample size was estimated using G*Power version 3.1.9.2 software program (Universität Kiel, Kiel, Germany) with power = 0.80, effect size (ES) [$\eta p^2 = 0.10$ (Effect size f = 0.33)], considering previous findings regarding the effect of prolonged cognitive effort on inhibitory control for professional soccer players (Gantois et al., 2020), $\alpha = 0.05$, number of experimental conditions = 4 (endurance vs. endurance + cognitive effort vs. endurance + cognitive effort + auditive distractor vs. endurance + auditive distractor), number of measurements = 3 (pre-experiment vs. 30-min post-experiment vs. 24-h follow-up), correlation among repeated measures = 0.5, and nonsphericity correction = 1. The results indicated that 20 subjects would be necessary for the study. Thus, 20 professional male soccer players (78.1 ± 6.9 kg, 1.77 ± 0.06 m, and 12.5 ± 5.3% body fat) were recruited utilizing a non-probabilistic method (i.e., recruitment by convenience). They were aged between 19 and 32 years old (M = 23.56 ± 3.8 years), played at a national level for at least five years (tenth-place in the last third division of the National Soccer League), and trained between five and six times a week for an average amount of 15.12 h (SD = 2.74) per week, and had been playing professional soccer for an average of 8.4 years (SD = 1.5). According to the Participant Classification Framework (McKay et al., 2022), this team soccer was classified as Tier 3 (Highly trained/National level). All participants had self-reported normal or corrected-to-normal vision and were unaware of the research objective. Before the experiment, the researchers informed the participants of the study's purpose, methods, and risks. All participants provided written informed consent before participation. This study was approved by the Ethics and Research for Humans Committee and followed the ethical principles in the Declaration of Helsinki.

2.2. Experimental design

The present study's design encompassed seven visits to the laboratory (one session for familiarization, two baseline visits, and four experimental conditions). The experimental sessions were performed in a randomized and counterbalanced crossover design [performed at the same time of the day (i.e., 8 h–10 h a.m.), under controlled temperature (~24 °C) and humidity (50–60%) conditions], separated by a one-week washout interval and maintaining the normal training routine. In the preliminary session, the soccer players were familiarized with a short version (10 trials) of the Stroop task, the multiple objects tracking task (MOT; 20 trials), auditive stimuli using headphones, and subjective scales. Then, they attended six more sessions. In the two baseline visits, the soccer players performed an incremental endurance test, incongruent Stroop task (100 trials) utilizing an eye-tracker to assess cognitive effort (i.e., pupil diameter), and MOT test (20 trials). These two sessions were designed to provide reliability measurements. In the experimental visits, the soccer players completed the incongruent Stroop task utilizing an eye-tracker to assess cognitive effort, MOT test, subjective mental fatigue, and heart rate variability before the endurance training and after 30-min and 24-h (i.e., follow-up) of recovery (see Figure 1). These sessions were designed to investigate the acute effect of prolonged cognitive effort (i.e., repeated MOT throughout the endurance training session) and/or an auditive distractor (i.e., constant crowd noise and coach's voice each 15 to 40-s, totalizing = 80 voices) on inhibitory control, MOT skill, HRV, and mental fatigue after a fixed endurance training session in professional soccer players.

2.3. Experimental conditions

2.3.1. Endurance session

The endurance training session was laboratory-based. Exercise intensity was 60%Δ of maximal aerobic velocity ($\dot{V}$ max) and performed on a motor-driven treadmill (model ATL, Inbramed, São Paulo, Brazil). The chosen exercise was intended to avoid early disengagement or exhaustion throughout the 40-min whole-body endurance task. Previous studies revealed that with intensity at 70%Δ Vmax, the participants remained less than 30-min in the task (Midgley et al., 2007; Walsh et al., 2010). The endurance training session was monitored with a heart rate monitor, a rating of perceived exertion, and cognitive RPE every 10 min. The endurance training session lasted 40-min.

2.3.2. Endurance and MOT session

The MOT task was performed simultaneously with endurance training. The participants performed the same endurance task on the treadmill and the MOT task (4 blocks of 20 trials of MOT) on the 55-inch screen in front of them. The subjects kept running intensity at the same workload throughout the MOT task. The verbal responses for the MOT task were available for the stimulus's response on the 55-inch screen.; *Endurance*, MOT, *and auditive distractor session*. The participants simultaneously performed the same endurance task and the same MOT task, while a headphone provided the auditive distractor (AD) stimuli (constant crowd noise and coach's voice each 15 to 40-s, totalizing = 80 voices). The subjects kept running intensity at the same workload throughout the MOT task. The verbal responses for the MOT task were available for the stimulus's response on the 55-inch screen.

2.3.3. Endurance and auditive distractor session

The participants performed the same endurance task, while a headphone provided the AD stimuli (constant crowd noise and coach's voice each 15 to 40-s, totalizing = 80 voices).

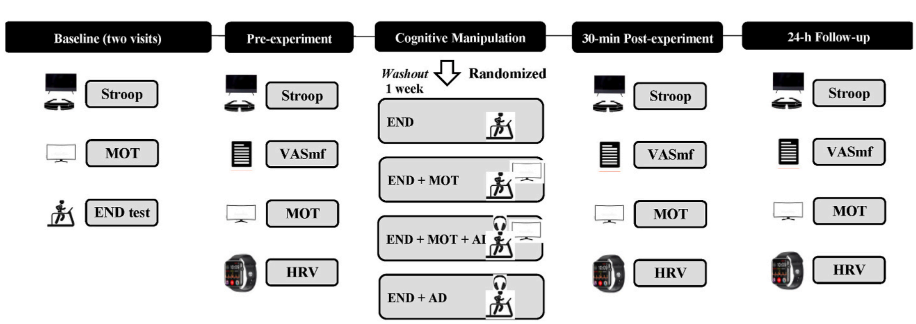

Figure 1. Experimental design of the study.
Note. MOT = multiple object tracking; END = endurance; VASmf = subjective mental fatigue; HRV = heart rate variability; AD = auditive distractor.

2.4. Measures

2.4.1. Incremental endurance test

The participants performed a maximal incremental test on a motor-driven treadmill (model ATL, Inbramed, São Paulo, Brazil). After a 3-min warm-up at 9 km/h, the speed was increased by 1 km/h every 3-min until exhaustion. Previous studies used this configuration for maximal incremental tests on a motor-driven treadmill (Bertuzzi et al., 2013; Machado et al., 2013). The participants received strong verbal encouragement (e.g., “go go go” and “do not stop”) to ensure the attainment of maximal values. These verbal encouragements were similar among participants. The highest velocity achieved during the test was recorded as the maximal aerobic velocity (Vmax). When the participants could not complete the entire last stage (<3 min), the Vmax was calculated using fractional time supported in the last stage multiplied by the increment rate.

2.4.2. Inhibitory control

The incongruent Stroop task was used to evaluate inhibitory control (PsychoPy v1.85.6, University of Nottingham, United Kingdom). We employed the Stroop task adopting 100 incongruent trials (one trial every 2500 ms). The task was performed in a silent and illuminated room, with the participants sitting comfortably on a chair in front of a 21 inches monitor and wearing an earphone to avoid noise distractions. In this task, four words (blue, yellow, red, and green) were presented in Arial font 60, once at a time, randomly, at the center of a computer screen (21 inches). The words were incongruently inked with the colors blue, yellow, red, or green. Subjects were instructed to press a colored button on the computer keyboard corresponding to the correct response as quickly and accurately as possible. The word’s ink color determined the correct response. If the ink was blue, green, or yellow, subjects should press the button corresponding to the ink color. If the ink color was red, the button that should be pressed was the button corresponding to the word’s meaning, not the ink color (e.g., if the word “blue” appeared inked in red, the button blue should be pressed) (Van Cutsem et al., 2018). Each word stayed on the screen for 2500 ms or until subjects pressed any answer button. The test was paced with an interval of 1000 ms between the response and a new stimulus. The behavioral performance was measured as accuracy (%) and response time (ms) for the answers. Two sessions with at least 72-h intervals were required to identify all participants’ Stroop task reliability values. The values found for accuracy and response time, respectively, were intraclass coefficient correlation (ICC) = 0.97 ($CI_{95\%}$ = 0.94 to 0.99) and ICC = 0.92 ($CI_{95\%}$ = 0.89 to 0.99).

2.4.3. Pupil diameter

Pupil diameter was recorded continuously throughout the Stroop color test with portable Eye Tracking-XG (Applied Science Laboratories, USA) equipment with a sampling frequency of 60 Hz. Gaze position was calibrated before the initial task using 3-point calibration. Using SMI BeGaze 3.2 software system (SMI, Berlin, Germany), the raw data of each participant was extracted to be further processed in RStudio software. A low-pass Butterworth filter (4 Hz) was applied to create a reliable signal profile. Then, the eye-tracker detected blinks, saccades events, artifacts, and outliers’ values (mean ± 2*SD) and removed them using a linear interpolation algorithm. We used pupil diameter to measure cognitive effort (Bafna & Hansen, 2021). We averaged the pupil diameter 500 ms before stimulus onset to measure resting pupil diameter. During this period, the participants saw a fixation cross with the same luminosity level as the letters, so there was no interference from eye reflexes to the environmental lighting. The stimulus-evoked pupil data were analyzed in RStudio software in the same way as the baseline pupil diameter. Resting pupil diameter and the mean pupil diameter peak activity for each time-on-task interval (i.e., mean each 30-s during the Stroop color task) were exported for further analysis.

2.4.4. Subjective mental fatigue

The subjective rating of mental fatigue was assessed using the 100 mm Visual Analogue Scale (VAS). This scale has two extremities anchored from 0 (none at all) to 100 (maximum). The participants were required to answer, “How mentally fatigued do you feel now?“. The definition of MF was provided to participants, and examples of “none at all” (no feelings of tiredness and lack of energy) and “maximum” (maximum feelings of tiredness and lack of energy) were explained based on prolonged tasks demanding cognitive activity. Participants were instructed to draw a single horizontal line to reflect MF throughout the 100 mm scale according to their perceived status. We measured the distance in millimeters from 0 to 100 indicated by the participant to quantify values.

2.4.5. Heart rate variability (HRV)

The R–R intervals recordings were assessed during 3-min in the seated position using a portable heart rate monitor HR with Bluetooth (H10, Polar Electro Oy®, Finland) with sampling at 1000Hz and downloaded via a validated smartphone app (Elite HRV app®) (Perrotta et al., 2017). The participants remained sitting for 5 min before initiating the resting HRV (Heart Rate Variability, 1996) in an air-conditioned room (24 °C). Data were analyzed in the last 2 min and after excluding the first 1 min to allow for heart rate stabilization. All the evaluations were performed under the same conditions. The analyzed variables were all R–R intervals, the success percentage of R–R interval differences more significant than 50 ms (pNN50), and the root means square of the successive differences (RMSSD). The R–R interval and RMSSD values were presented in milliseconds (ms).

2.4.6. Multiple object tracking (MOT)

The MOT is software under the NeuroTracker™ system licensed by the University of Montreal (CogniSens Inc.; Montreal, QC, Canada) that measures visual tracking threefold speed and score (percentage of accuracy). The MOT task was chosen because it has been proposed as an optimal training procedure for isolating critical mental abilities when processing dynamic scenes, such as during sports activities (Faubert, 2013; Faubert & Sidebottom, 2012). The ability to track multiple elements (including teammates, opponents, and the ball or puck) is a capacity that is highly desirable in team sports and, in particular, for soccer. Previous studies also used the MOT task to evaluate the ability to track multiple elements in team sports players (Romeas et al., 2016, 2019). In the first phase, eight balls were presented, and four of these eight balls were highlighted in a different color for 1 s and then returned to their original color. In the next phase, the balls moved at a defined speed in space for 8 s. The spheres followed a linear trajectory in the virtual space. Deviation occurred only when the balls collided with each other or the walls. The participant’s task was to track the four initially color-marked balls and then, when they came to a stop, to give a verbal response that identified the four balls illuminated originally at the beginning of the experiment by their number. This was followed by a feedback phase in which the participant named the balls’ numbers, and the subsequent trial began. Participants had to keep their gaze centered and peripherally follow the balls (Samsung, 55″ inches). After each correct response, the dependent variable (speedball displacement) was increased by 0.05 cm s^{-1}, and it was decreased by the same proportion after each incorrect response, resulting in a threshold criterion of 50%. Twenty trials were performed at the start (pre-experiment), 30-min post-experiment, and 24-h follow-up. Each session lasted about 8 min. The intraclass coefficient correlation (I_{CC}) was used to determine the reliability of MOT threefold speed (ICC = 0.75, $CI_{95\%}$ = 0.71 to 0.84) and score (I_{CC} = 0.80, $CI_{95\%}$ = 0.74 to 0.89) in baseline visits.

2.4.7. Rating of perceived exertion (RPE)

RPE was monitored throughout endurance training sessions using a CR-10 scale (Foster et al., 2001). Specifically, the participants were instructed to provide their RPE every 10 min. So, according to the experimental condition, we compared RPE for 10, 20, 30, and 40 min for time throughout the endurance training session. Also, the session-RPE quantified the internal training load (Foster et al., 2001) measured 30-min after the endurance training session. The participant was asked to demonstrate their intensity perception of the session from the 10-point Borg scale (0 = rest to 10 = maximum effort) according to the method developed by Foster et al. (2001). The product of the values demonstrated by the RPE scale and the total time in minutes of the session [i.e., only 40-min of duration for experimental sessions without considering the warm-up and cool-down (30-min after the experimental session)] was calculated and displayed in arbitrary units (a.u.). Athletes were familiarized with the session-RPE method for 30 days before beginning the investigation.

2.4.8. Cognitive rating perceived exertion (cognitive-RPE)

The cognitive RPE was monitored throughout endurance training sessions using a CR-10 scale (Foster et al., 2001), varying from no exertion (0) to maximum effort (10), which is the one that answers the question “How much mental effort and decision-making has this task required?” (Fuster et al., 2021). Specifically, the participants were instructed to provide their cognitive RPE every 10 min. So, according to the experimental condition, we compared cognitive RPE for 10, 20, 30, and 40 min for time throughout the endurance training session. In addition, the cognitive RPE for the experimental session was quantified by the session-cognitive-RPE (Foster et al., 2001) measured 30-min after the endurance training session. The participant was asked to demonstrate the cognitive demand perception of the session from the 10-point Borg scale (0 = no exertion to 10 = maximum effort) according to the method developed by Farrow et al. (2008). The participants were familiarized with the session-cognitive-RPE method for 30 days before beginning the investigation.

2.5. Statistical analysis

A two-way ANOVA with repeated measures was used to compare the inhibitory control (accuracy, response time, and pupil diameter), subjective mental fatigue, HRV (R–R intervals, pNN50, and RMSSD), and MOT (score and threefold speed) using the group (endurance vs. endurance + MOT vs. endurance + MOT + AD vs. endurance + AD) and time (pre-experiment vs. 30-min post-experiment vs. 24-h follow-up) as fixed factors and subjects as the random factor. A two-way ANOVA with repeated measures also was used to analyze a group (endurance vs. endurance + MOT vs. endurance + MOT + AD vs. endurance + AD) x time (1st 10-min vs. 2nd 10-min vs. 3rd 10-min vs. 4th 10-min) interaction for RPE, cognitive-RPE, and heart rate throughout the training session, adopting group and time as fixed factors and subjects as the random factor. A Bonferroni post hoc test was used to identify possible statistical differences. Partial eta squared (ηp^2) was used to determine the effect size (ES) and was interpreted using the following cut-off points (Cohen, 1988): small effect, $\eta p^2 < 0.03$; moderate effect, $0.03 \leq \eta p^2 < 0.10$; large effect, $0.10 \leq \eta p^2 < 0.20$; very large effect, $\eta p^2 \geq 0.20$. The Mauchly test assessed the sphericity assumption, and Greenhouse-Geiser correction was used when needed. A one-way ANOVA with repeated measures was utilized to compare internal training load and cognitive-RPE training sessions between experimental conditions (endurance vs. endurance + MOT vs. endurance + MOT + AD vs. endurance + AD). A Bonferroni post hoc test was used to identify possible statistical differences. Data were processed in the Statistical Package for Social Sciences Version 21.0 (IBM Corp., Armonk, NY, USA) and GraphPad Prism 8 (San Diego, CA, USA). A significance level of 5% was adopted.

3. Results

There was no significant effect of experimental conditions order ($F_{(2.709, 51)} = 0.34$; $p = 0.86$; $\eta p^2 = 0.01$).

3.1. Inhibitory control

3.1.1. Accuracy

There was no main effect of condition for accuracy (Supplemental material; $F_{(4, 76)} = 0.94$; $p = 0.87$; $\eta p^2 = 0.001$). Also, there was no main effect of time for accuracy ($F_{(3, 57)} = 1.33$; $p = 0.72$; $\eta p^2 = 0.001$). There was no condition × time interaction for accuracy ($F_{(12, 228)} = 0.60$; $p = 0.93$; $\eta p^2 = 0.001$).

3.1.2. Response time

There was a significant condition × time interaction for response time (Figure 2A; $F_{(12, 228)} = 37.29$; $p = 0.001$; $\eta p^2 = 0.16$). A higher response time was found for the endurance + MOT + AD and endurance + MOT than the other experimental conditions ($p < 0.05$).

3.1.3. Pupil diameter

There was a significant condition × time interaction for pupil diameter (Figure 2B; $F_{(12, 228)} = 9.68$; $p = 0.01$; $\eta p^2 = 0.08$). A lower pupil diameter was found for the endurance + MOT + AD and endurance + MOT than the other experimental conditions ($p < 0.05$).

3.2. Subjective mental fatigue

There was a significant condition × time interaction for subjective mental fatigue (Figure 3A; $F_{(12, 228)} = 8.62$; $p = 0.001$; $\eta p^2 = 0.46$). A higher subjective mental fatigue was found for the endurance + MOT + AD and endurance + MOT than the other experimental conditions ($p < 0.05$). Post-hoc test showed higher subjective mental fatigue in 30-min post-experiment and follow-up 24-h than pre-experiment ($p < 0.05$).

3.3. HRV

3.3.1. R–R intervals

There was no main effect of condition for R–R intervals (Figure 3B; $F_{(4, 76)} = 1.42$; $p = 0.34$; $\eta p^2 = 0.02$). There was a significant main effect of time for R–R intervals ($F_{(3, 57)} = 237.30$; $p = 0.001$; $\eta p^2 = 0.24$). Post-hoc test showed lower R–R intervals in 30-min post-experiment than pre-experiment ($p < 0.05$). There was no condition × time interaction for R–R intervals ($F_{(12, 228)} = 0.20$; $p = 0.76$; $\eta p^2 = 0.001$).

3.3.2. RMSSD

There was no main effect of condition for RMSSD (Figure 3C; $F_{(4, 76)} = 0.95$; $p = 0.41$; $\eta p^2 = 0.02$). There was a significant main effect of time for RMSSD ($F_{(3, 57)} = 195.30$; $p = 0.001$; $\eta p^2 = 0.28$). Post-hoc test showed lower RMSSD in 30-min post-experiment than pre-experiment ($p < 0.05$). There was no condition × time interaction for RMSSD ($F_{(12, 228)} = 0.92$; $p = 0.43$; $\eta p^2 = 0.001$).

3.3.3. pNN50

There was no main effect of condition for pNN50 (Figure 3D; $F_{(4, 76)} = 2.82$; $p = 0.07$; $\eta p^2 = 0.03$). There was a significant main effect of time for pNN50 ($F_{(3, 57)} = 234.10$; $p = 0.001$; $\eta p^2 = 0.30$). Post-hoc test showed lower pNN50 in 30-min post-experiment than pre-experiment ($p < 0.05$). There was no condition × time interaction for R–R intervals ($F_{(12, 228)} = 0.60$; $p = 0.58$; $\eta p^2 = 0.001$).

3.4. MOT

3.4.1. Score

There was a significant condition × time interaction for score (Figure 2C; $F_{(12, 228)} = 17.24$; $p = 0.001$; $\eta p^2 = 0.23$). A lower score was found for the endurance + MOT + AD and endurance + MOT than in the other experimental conditions ($p < 0.05$).

3.4.2. Threshold speed

There was a significant condition × time interaction for threshold

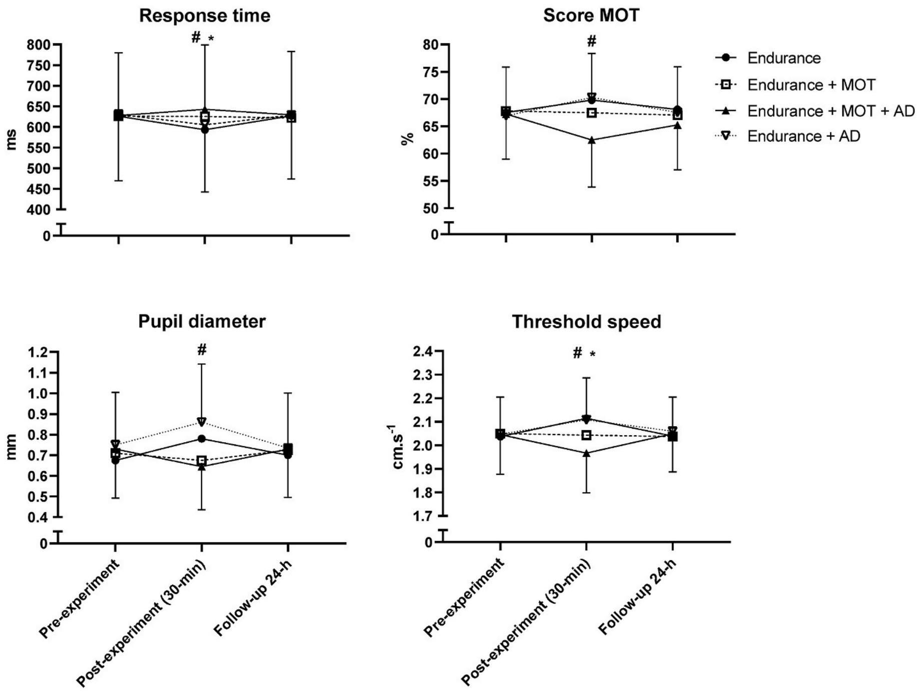

Figure 2. Stroop response time (A), pupil diameter throughout Stroop (B), score MOT (C), and threshold speed for MOT (D) according to experimental conditions and time.
Note. MOT = multiple object tracking; AD = auditive distractor; *p < 0.05 difference for pre-experiment; #p < 0.05 difference between Endurance + MOT + AD and Endurance + MOT than other experimental conditions.

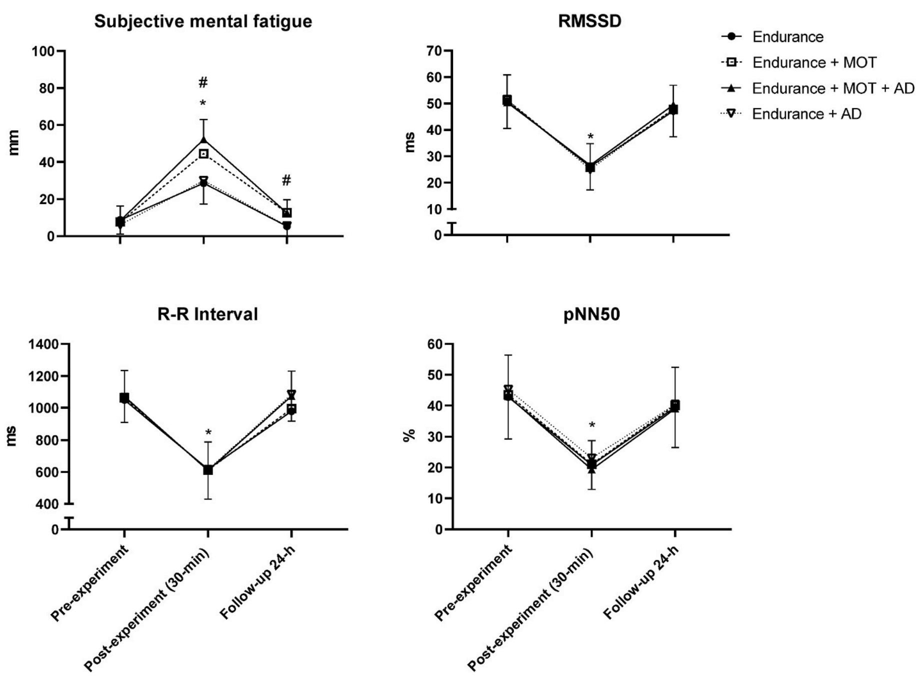

Figure 3. Subjective mental fatigue (A) and HRV [R–R interval (B), RMSSD (C), and pNN50 (D)] according to experimental conditions and time.
Note. RMSSD = root mean square of the successive differences; pNN50 = R–R interval differences greater than 50 ms; MOT = multiple object tracking; AD = auditive distractor; *p < 0.05 difference for pre-experiment; #p < 0.05 difference between Endurance + MOT + Distractor and Endurance + MOT than other experimental conditions.

speed (Figure 2D; $F_{(12,\ 228)} = 31.85$; $p = 0.001$; $\eta p^2 = 0.29$). A lower threshold speed was found for the endurance + MOT + AD and endurance + MOT than in the other experimental conditions ($p < 0.05$).

3.5. RPE

There was a significant main effect of condition for RPE (Figure 4A; $F_{(4,\ 76)} = 99.73$; $p = 0.001$; $\eta p^2 = 0.16$). A higher RPE was found for the endurance + MOT + AD than in the other experimental conditions ($p < 0.05$). Also, there was a significant main effect of time for RPE ($F_{(4,\ 76)} = 121.02$; $p = 0.001$; $\eta p^2 = 0.65$). Post-hoc tests showed increased RPE throughout the training session ($p < 0.05$). There was no condition × time interaction for RPE ($F_{(16,\ 224)} = 0.22$; $p = 0.99$; $\eta p^2 = 0.001$).

3.5.1. Internal training load

There was a significant difference in internal training load (Figure 5A; $F_{(4,\ 76)} = 24.38$; $p = 0.001$). A higher internal training load was found for the endurance + MOT + AD than the other experimental conditions ($p < 0.05$).

3.6. Cognitive-RPE

There was a significant condition × time interaction for cognitive-RPE (Figure 4B; $F_{(16,\ 224)} = 6.01$; $p = 0.001$; $\eta p^2 = 0.17$). A higher

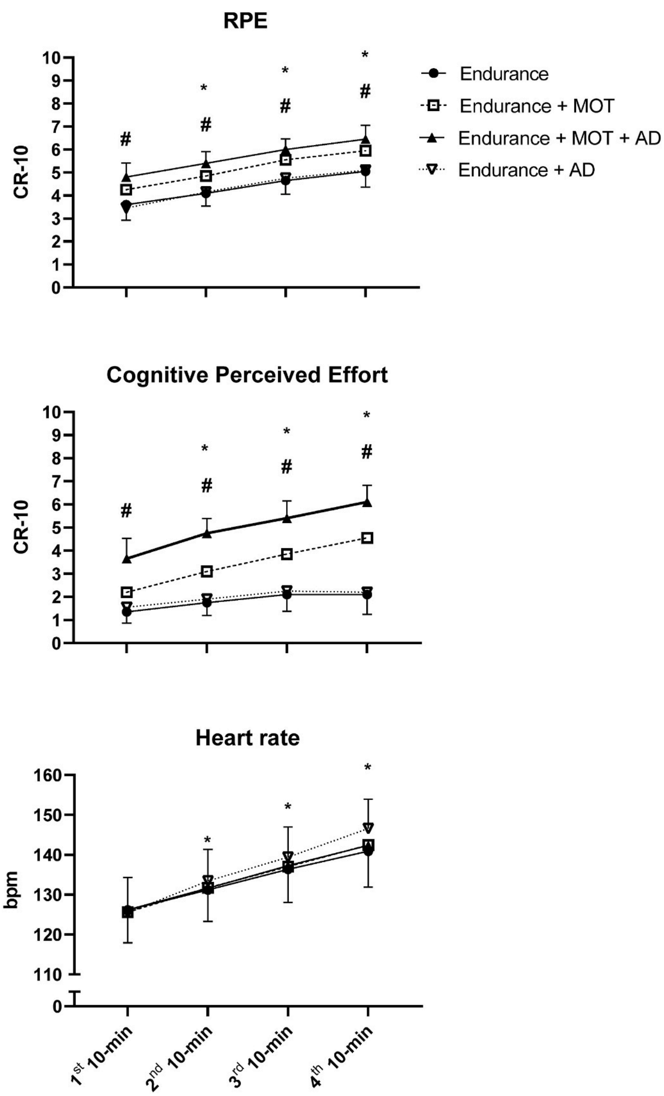

Figure 4. RPE (A), cognitive-RPE (B), and heart rate (C) throughout training session according to experimental conditions.
Note. MOT = multiple object tracking; AD = auditive distractor; RPE = rating perceived exertion. *p < 0.05 difference for 1st 10-min; #p < 0.05 difference between Endurance + MOT + AD and other experimental conditions.

cognitive RPE was found for the endurance + MOT + AD than in the other experimental conditions ($p < 0.05$).

3.6.1. *Cognitive-RPE training session*

The cognitive-RPE training session significantly differed (Figure 5B; $F_{(4,\ 76)} = 73.72$; $p = 0.001$). A higher cognitive RPE for the training session was found for the endurance + MOT + AD than in the other experimental conditions ($p < 0.05$). Also, a higher cognitive RPE for the training session was found for the endurance + MOT than for the endurance and endurance + AD experimental conditions ($p < 0.05$).

3.7. *Heart rate*

There was no main effect of condition on heart rate (Figure 4C; $F_{(4,\ 76)} = 1.54$; $p = 0.20$; $\eta p^2 = 0.02$). There was a significant main effect of time for heart rate ($F_{(4,\ 76)} = 66.28$; $p = 0.001$; $\eta p^2 = 0.47$). Post-hoc tests showed increased heart rate throughout the training session ($p < 0.05$). There was no condition × time interaction for heart rate ($F_{(16,\ 224)} = 0.38$; $p = 0.94$; $\eta p^2 = 0.001$).

4. Discussion

The objective of this study was to analyze the acute effect of cognitive effort and auditive distractor with 24-h follow-up throughout a prolonged endurance session on inhibitory control, subjective mental fatigue, MOT skill, and HRV in professional soccer players. The main findings indicated impaired inhibitory control and increased mental fatigue 30-min after the endurance training session with cognitive effort and the auditive distractor, returning baseline values after a 24-h follow-up. These results partially corroborate the initial hypothesis.

The results regarding inhibitory control revealed that only response time but not accuracy was impaired when endurance training sessions with cognitive effort had or did not have an auditive distractor. Thus, it is possible to suggest that an endurance session combined with highly demanding cognitive effort could overload the interference control cognitive system, although this effect is dissipated in the posterior 24-h of recovery. The reduction of cognitive effort invested (i.e., attenuated pupil diameter in 30-min post-experiment) after the endurance + MOT + AD and endurance + MOT experimental sessions can explain the impaired response time for the inhibitory control task (i.e., throughout the Stroop task). These results corroborate previous findings. Herlambang et al. (2021) also found reduced cognitive effort (i.e., attenuated pupil diameter) after prolonged inhibitory control tasks. Notably, performance decline due to fatigue may depend on a reduced desire to exert further cognitive effort, suggesting that under mental fatigue, the integrated effort-value function would be shifted toward a diminished preference for cognitive effort (Kok, 2022).

Regarding the experimental design of the present study, our test attempted to simulate a soccer training session in terms of cognitive

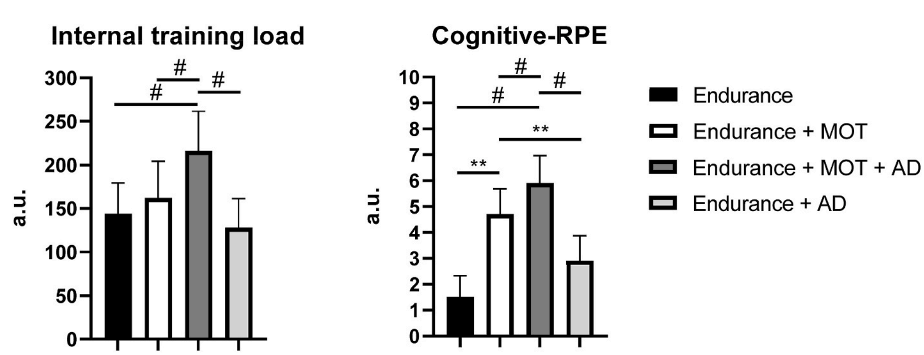

Figure 5. Internal training load (A) and cognitive-RPE training session (B) according to experimental conditions
Note. MOT = multiple object tracking; AD = auditive distractor; RPE = rating perceived exertion; **p < 0.05 difference between Endurance + MOT than Endurance and Endurance + AD; #p < 0.05 difference between Endurance + MOT + AD and other experimental condition.

load. During a soccer training session, the players must frequently make endurance activities with cognitive tasks (e.g., MOT, decision-making, anticipation, and visuomotor), which could overload the interference control cognitive system throughout the training session and impair executive functions (Bigliassi, 2021). However, it is important to highlight that in our experiment, the endurance session was only 40-min, less than half compared to a training session commonly adopted by professional soccer coaches (e.g., superior to 90-min). According to our results, auditive distractors did not add to the cognitive overload. One possible explanation is that professional soccer players are well-trained for physical endurance, which seems to be related to higher inhibitory control levels. Previous findings revealed that well-trained athletes for physical endurance showed better inhibitory control than amateur athletes (Martin et al., 2016). A relationship exists between high inhibitory control, high physical endurance capacity, and the ability to inhibit auditive distractors (Bigliassi, 2021). Thus, the professional soccer players in the present study might be well-adapted to inhibit auditive distractors during soccer training sessions due to high inhibitory control (e.g., see the Stroop task response time in Figure 2). Another explanation for the absence of cognitive overload during auditive distractors could be the whole-body physical task used in the present investigation, which was not similar to the intermittent nature of the physiological demands in soccer training sessions. Then, more studies should be conducted to clarify the effect of auditive distractor and cognitive effort on inhibitory control in professional team sports players.

The results showed increased subjective mental fatigue for all experimental conditions, including when athletes only ran. Previous scientific investigations showed that whole-body endurance tasks could cause mental fatigue (Pereira et al., 2021; Xu et al., 2018), corroborating the experimental endurance session findings. The subjective mental fatigue may increase during prolonged whole-body endurance tasks when combined with a cognitive task (Dallaway et al., 2021, 2023). The findings in the present study revealed increased subjective mental fatigue following 30-min and until 24-h after the endurance exercise with cognitive effort (i.e., endurance + MOT and endurance + MOT + AD) than other experimental conditions with or without auditive distractor (i.e., endurance and endurance + AD). It might be explained by greater cognitive effort for dual tasks using the cognitive interference control system, for example, prolonged physical endurance (De Wachter et al., 2021; Dellaway et al., 2021) and MOT (Faubert & Sidebottom, 2012) together. The findings for subjective mental fatigue in the present study showed no difference between endurance exercise plus cognitive effort with or without auditive distractor, refuting the initial hypothesis in which we indicated that the auditive distractor, for example, the presence of journalist and fans during training sessions, would increase the demands on the interference control cognitive and would consequently cause increased mental fatigue in soccer players. Perhaps the interference control cognitive system can have limited resources and may not increase the mental fatigue degree when other tasks also demand cognitive resources, such as endurance and MOT. Corroborating this statement, previous studies that combined different stressors (e.g., mental, physical, or environmental) also found no increase in mental fatigue (Alder et al., 2021; Valenza et al., 2020). After 24-h follow-up soccer players still presented greater subjective mental fatigue at endurance + MOT and endurance + MOT + AD experimental conditions when compared to the other experimental conditions. A greater mental fatigue after high cognitive load seems to affect the time course for mental recovery. Previous findings demonstrated cumulated subjective mental fatigue after repeated prolonged cognitive effort immediately prior physical training sessions for high-level volleyball (Fortes, Berriel et al., 2022) and swimming athletes (Fortes, Nakamura, Lima-Junior, Ferreira, & Fonseca, 2022). Also, Russel et al. (2022b) demonstrated cumulated subjective mental fatigue throughout 16-weeks pre-season in elite netball players, while Russel et al. (2022a) showed that perceived measures of mental fatigue increased toward the later part of the elite netball season. These body of evidence indicates that the time course for mental fatigue recovery can increase depending on the cognitive load throughout the training sessions (Loch et al., 2020). It is necessary to conduct more studies about the time-course of mental fatigue recovery in team sports players utilizing different cognitive loads throughout the training sessions.

Regarding HRV, the findings revealed attenuation for all experimental conditions after the training sessions. These results corroborate studies with whole-body endurance tasks (Leicht et al., 2018; Nakamura et al., 2016). Previous studies demonstrated impaired HRV throughout highly demanding cognitive tasks (Kim et al., 2018; Qin et al., 2021; Tanaka et al., 2009, 2012), a finding not further supported by our results. It is important to highlight that the HRV measure was made throughout the cognitive task session in previous studies, while in the present study, HRV was measured immediately before, 30-min, and 24-h after the training session in the rest state. Considering previous scientific investigations that measured HRV immediately prior and after prolonged cognitive task (Fortes et al., 2019; Gantois et al., 2020) found no changes in HRV, which corroborates the findings of the present study. So, these differences in methods regarding the moment for HRV measure can explain the difference between findings about the effect of high cognitive load on HRV. Although the reduction of HRV can have a relationship with impaired recovery (Bellenger et al., 2016), our study showed that a 24-h period was sufficient for full recovery of the autonomic nervous system function.

Regarding MOT skill, the results revealed a lower score and threshold speed when soccer players ran with and without AD. As demonstrated in previous studies, prolonged cognitive effort impairs performance during cognitive tasks (Gantois et al., 2020; Smith et al., 2016; Van-Cutsem et al., 2022). Only a 40-min endurance training session with a cognitive task was sufficient for the impairment threshold speed in MOT skill after the session. Furthermore, repeated MOT tasks (e.g., throughout the 40-min) require sustained peripheral attention, increasing the attentional load (Faubert, 2013), and probably increasing the cognitive load. This study's cognitive RPE findings corroborate this hypothesis (i.e., endurance + MOT and endurance + MOT + AD). Thus, greater cognitive RPE during the MOT task might explain the score and threshold speed after 30-min for the endurance + MOT and endurance + MOT + AD. In addition, regarding AD, the MOT skill can be even more impaired combined with previous cognitive effort due to interference control cognitive system overload. However, the findings refute this hypothesis. It would suggest that soccer players can deal with additional cognitive stress or add cognitive resources to compensate for the overload of the interference control system. According to Bigliassi (2021), sensory distractions (e.g., auditive distractors) may potentially increase activation in the brain and protect the organism against the detrimental effects of fatigue, which explain the findings regarding AD on MOT skill. This neural mechanism might be responsible for improving cognitive task performance consistently reported when individuals exercise in the presence of environmental distractions (Bigliassi et al., 2016, 2017).

Considering RPE in the training session, the values increased throughout the endurance session with higher values for running with cognitive activity and the AD than in the other experimental conditions. Similar findings were also found for internal training load and cognitive-RPE (i.e., higher END + MOT + Distractor values than in the other experimental conditions). It can be explained by increasing cognitive effort when there is a demand for greater interference in cognitive control throughout the task (Bigliassi, 2021). It is relevant to note that auditive sensory signals usually elicit multifarious effects on attentional control (Yashiro et al., 2019). This is mainly because real-life situations are representative of complex biopsychosocial interactions; therefore, sensory signals can elicit numerous perceptual responses (Bigliassi, 2021), for example, changes in RPE and cognitive-RPE, throughout whole-body endurance tasks with an AD.

The findings from the present study are original and bring important new scientific knowledges. The MOT task and whole-body endurance increased perceived cognitive effort throughout the 40-min session and increased mental fatigue until 24-h following the session. These findings could indicate to soccer coaches and athletes that for training sessions in which soccer players track relevant stimuli (e.g., teammates, opponents, goalkeeper, ball, and empty spaces) and run at moderate intensity (e.g., 60% of maximum aerobic velocity), they might need greater interval recovery between training sessions or the next training session should be prescribed with a lower cognitive and physical demand.

'Despite the originality and unheard findings, this scientific investigation presented some limitations. It was not performed with neuro-imaging (e.g., fMRI) or brain oxygenation (e.g., fNIRS) measured throughout experimental sessions. Also, only professional soccer players were participants in this study. The physical task (40-min of whole-body endurance) does not thoroughly represent the intermittent nature of the physiological demands in soccer. Another limitation was the duration of the experimental session, which was lower than a soccer training session. So, we cannot claim that the changes we observed are directly translatable to a real-life soccer game. However, this study indicates changes that might be highly beneficial when applied to mechanistic knowledge.

5. Conclusion

From this study, we conclude that cognitive effort throughout a prolonged endurance session impaired inhibitory control and increased subjective mental fatigue without promoting greater MOT skill and HRV changes in professional soccer players. However, considering the AD, the findings showed no additional effect on these parameters. It is reasonable to suggest that professional soccer players frequently perform brain endurance training sessions to inhibit impaired executive functions and increased mental fatigue after training sessions. It is recommended that sports scientists analyze whether AD might increase cognitive effort during a prolonged endurance task in amateur and recreational soccer players. Finally, it is suggested that sports scientists analyze how long athletes recover from mental fatigue following a training session or simulated soccer match with different cognitive and physical loads.

Declaration of competing interest

None

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.psychsport.2023.102533.

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